OPERATING SYSTEMS – WEEK 8

Page Replacement Algorithms – Questions and Step-by-Step Answers

QUESTION 1

Question: For the following sequence of pages find the hit and miss ratio using FIFO, LRU, MRU and Optimal Page Replacement technique with physical memory size 3.

Page sequence: 3 2 1 3 4 1 6 2 4 3 4 2 1 4 5 2 1 3 4
Number of references: 19 **Number of frames:** 3

Formulas:
Hit Ratio = Hits / Total References × 100
Miss Ratio = Page Faults / Total References × 100
Page Faults = Total References - Hits

QUESTION 1 – FIFO

FIFO – 3 Frames: Step-by-Step Calculation

Total references = 19
Hits = 6
Page faults = 19 - 6 = 13
Hit ratio = 6/19 × 100 = 31.58%
Miss ratio = 13/19 × 100 = 68.42%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	3	3	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	3	2	-	-	F	an empty frame is available, so the page is loaded.
3	1	3	2	1	-	F	an empty frame is available, so the page is loaded.
4	3	3	2	1	-	H	page is already present in memory.
5	4	4	2	1	-	F	replace 3 because it entered memory first (FIFO).
6	1	4	2	1	-	H	page is already present in memory.
7	6	4	6	1	-	F	replace 2 because it entered memory first (FIFO).
8	2	4	6	2	-	F	replace 1 because it entered memory first (FIFO).
9	4	4	6	2	-	H	page is already present in memory.
10	3	3	6	2	-	F	replace 4 because it entered memory first (FIFO).
11	4	3	4	2	-	F	replace 6 because it entered memory first (FIFO).
12	2	3	4	2	-	H	page is already present in memory.
13	1	3	4	1	-	F	replace 2 because it entered memory first (FIFO).
14	4	3	4	1	-	H	page is already present in memory.
15	5	5	4	1	-	F	replace 3 because it entered memory first (FIFO).
16	2	5	2	1	-	F	replace 4 because it entered memory first (FIFO).
17	1	5	2	1	-	H	page is already present in memory.
18	3	5	2	3	-	F	replace 1 because it entered memory first (FIFO).
19	4	4	2	3	-	F	replace 5 because it entered memory first (FIFO).

Final calculation: Page faults = 19 - 6 = 13. Hit ratio = 6/19 × 100 = 31.58%. Miss ratio = 13/19 × 100 = 68.42%.

QUESTION 1 – LRU

LRU – 3 Frames: Step-by-Step Calculation

Total references = 19
Hits = 5
Page faults = 19 - 5 = 14
Hit ratio = 5/19 × 100 = 26.32%
Miss ratio = 14/19 × 100 = 73.68%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	3	3	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	3	2	-	-	F	an empty frame is available, so the page is loaded.
3	1	3	2	1	-	F	an empty frame is available, so the page is loaded.
4	3	3	2	1	-	H	page is already present in memory.
5	4	3	4	1	-	F	replace 2 because it was used least recently (LRU).
6	1	3	4	1	-	H	page is already present in memory.
7	6	6	4	1	-	F	replace 3 because it was used least recently (LRU).
8	2	6	2	1	-	F	replace 4 because it was used least recently (LRU).
9	4	6	2	4	-	F	replace 1 because it was used least recently (LRU).
10	3	3	2	4	-	F	replace 6 because it was used least recently (LRU).
11	4	3	2	4	-	H	page is already present in memory.
12	2	3	2	4	-	H	page is already present in memory.
13	1	1	2	4	-	F	replace 3 because it was used least recently (LRU).
14	4	1	2	4	-	H	page is already present in memory.
15	5	1	5	4	-	F	replace 2 because it was used least recently (LRU).
16	2	2	5	4	-	F	replace 1 because it was used least recently (LRU).
17	1	2	5	1	-	F	replace 4 because it was used least recently (LRU).
18	3	2	3	1	-	F	replace 5 because it was used least recently (LRU).
19	4	4	3	1	-	F	replace 2 because it was used least recently (LRU).

Final calculation: Page faults = 19 - 5 = 14. Hit ratio = 5/19 × 100 = 26.32%. Miss ratio = 14/19 × 100 = 73.68%.

QUESTION 1 – MRU

MRU – 3 Frames: Step-by-Step Calculation

Total references = 19
Hits = 7
Page faults = 19 - 7 = 12
Hit ratio = 7/19 × 100 = 36.84%
Miss ratio = 12/19 × 100 = 63.16%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	3	3	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	3	2	-	-	F	an empty frame is available, so the page is loaded.
3	1	3	2	1	-	F	an empty frame is available, so the page is loaded.
4	3	3	2	1	-	H	page is already present in memory.
5	4	4	2	1	-	F	replace 3 because it was used most recently (MRU).
6	1	4	2	1	-	H	page is already present in memory.
7	6	4	2	6	-	F	replace 1 because it was used most recently (MRU).
8	2	4	2	6	-	H	page is already present in memory.
9	4	4	2	6	-	H	page is already present in memory.
10	3	3	2	6	-	F	replace 4 because it was used most recently (MRU).
11	4	4	2	6	-	F	replace 3 because it was used most recently (MRU).
12	2	4	2	6	-	H	page is already present in memory.
13	1	4	1	6	-	F	replace 2 because it was used most recently (MRU).
14	4	4	1	6	-	H	page is already present in memory.
15	5	5	1	6	-	F	replace 4 because it was used most recently (MRU).
16	2	2	1	6	-	F	replace 5 because it was used most recently (MRU).
17	1	2	1	6	-	H	page is already present in memory.
18	3	2	3	6	-	F	replace 1 because it was used most recently (MRU).
19	4	2	4	6	-	F	replace 3 because it was used most recently (MRU).

Final calculation: Page faults = 19 - 7 = 12. Hit ratio = 7/19 × 100 = 36.84%. Miss ratio = 12/19 × 100 = 63.16%.

QUESTION 1 – Optimal

Optimal – 3 Frames: Step-by-Step Calculation

Total references = 19
Hits = 9
Page faults = 19 - 9 = 10
Hit ratio = 9/19 × 100 = 47.37%
Miss ratio = 10/19 × 100 = 52.63%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	3	3	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	3	2	-	-	F	an empty frame is available, so the page is loaded.
3	1	3	2	1	-	F	an empty frame is available, so the page is loaded.
4	3	3	2	1	-	H	page is already present in memory.
5	4	4	2	1	-	F	replace 3 because its next use is farthest in the future (Optimal).
6	1	4	2	1	-	H	page is already present in memory.
7	6	4	2	6	-	F	replace 1 because its next use is farthest in the future (Optimal).
8	2	4	2	6	-	H	page is already present in memory.
9	4	4	2	6	-	H	page is already present in memory.
10	3	4	2	3	-	F	replace 6 because it is not used again in the future (Optimal).
11	4	4	2	3	-	H	page is already present in memory.
12	2	4	2	3	-	H	page is already present in memory.
13	1	4	2	1	-	F	replace 3 because its next use is farthest in the future (Optimal).
14	4	4	2	1	-	H	page is already present in memory.
15	5	5	2	1	-	F	replace 4 because its next use is farthest in the future (Optimal).
16	2	5	2	1	-	H	page is already present in memory.
17	1	5	2	1	-	H	page is already present in memory.
18	3	3	2	1	-	F	replace 5 because it is not used again in the future (Optimal).
19	4	4	2	1	-	F	replace 3 because it is not used again in the future (Optimal).

Final calculation: Page faults = 19 - 9 = 10. Hit ratio = 9/19 × 100 = 47.37%. Miss ratio = 10/19 × 100 = 52.63%.

QUESTION 1 – FINAL ANSWER

Algorithm	Hits	Calculation of Page Faults	Page Faults	Hit Ratio	Miss Ratio
FIFO	6	19 - 6 = 13	13	6/19 × 100 = 31.58%	13/19 × 100 = 68.42%
LRU	5	19 - 5 = 14	14	5/19 × 100 = 26.32%	14/19 × 100 = 73.68%
MRU	7	19 - 7 = 12	12	7/19 × 100 = 36.84%	12/19 × 100 = 63.16%
Optimal	9	19 - 9 = 10	10	9/19 × 100 = 47.37%	10/19 × 100 = 52.63%

QUESTION 2

Question: Apply FIFO Page Replacement Algorithm for the following sequence of pages and check whether Belady's anomaly exists or not. For checking Belady's anomaly take physical memory size as 3 or 4 frames. Also apply the other page replacement algorithms, calculate efficiency using hit ratio and miss ratio, and identify the suitable algorithm for this question.

Page sequence: 1 2 3 4 1 2 5 1 2 3 4 5
Number of references: 12

QUESTION 2 – 3 FRAMES

FIFO

FIFO – 3 Frames: Step-by-Step Calculation

Total references = 12
Hits = 3
Page faults = 12 - 3 = 9
Hit ratio = 3/12 × 100 = 25.00%
Miss ratio = 9/12 × 100 = 75.00%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	4	2	3	-	F	replace 1 because it entered memory first (FIFO).
5	1	4	1	3	-	F	replace 2 because it entered memory first (FIFO).
6	2	4	1	2	-	F	replace 3 because it entered memory first (FIFO).
7	5	5	1	2	-	F	replace 4 because it entered memory first (FIFO).
8	1	5	1	2	-	H	page is already present in memory.
9	2	5	1	2	-	H	page is already present in memory.
10	3	5	3	2	-	F	replace 1 because it entered memory first (FIFO).
11	4	5	3	4	-	F	replace 2 because it entered memory first (FIFO).
12	5	5	3	4	-	H	page is already present in memory.

Final calculation: Page faults = 12 - 3 = 9. Hit ratio = 3/12 × 100 = 25.00%. Miss ratio = 9/12 × 100 = 75.00%.

LRU

LRU – 3 Frames: Step-by-Step Calculation

Total references = 12
Hits = 2
Page faults = 12 - 2 = 10
Hit ratio = 2/12 × 100 = 16.67%
Miss ratio = 10/12 × 100 = 83.33%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	4	2	3	-	F	replace 1 because it was used least recently (LRU).
5	1	4	1	3	-	F	replace 2 because it was used least recently (LRU).
6	2	4	1	2	-	F	replace 3 because it was used least recently (LRU).
7	5	5	1	2	-	F	replace 4 because it was used least recently (LRU).
8	1	5	1	2	-	H	page is already present in memory.
9	2	5	1	2	-	H	page is already present in memory.
10	3	3	1	2	-	F	replace 5 because it was used least recently (LRU).
11	4	3	4	2	-	F	replace 1 because it was used least recently (LRU).
12	5	3	4	5	-	F	replace 2 because it was used least recently (LRU).

Final calculation: Page faults = 12 - 2 = 10. Hit ratio = 2/12 × 100 = 16.67%. Miss ratio = 10/12 × 100 = 83.33%.

MRU

MRU – 3 Frames: Step-by-Step Calculation

Total references = 12
Hits = 5
Page faults = 12 - 5 = 7
Hit ratio = 5/12 × 100 = 41.67%
Miss ratio = 7/12 × 100 = 58.33%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	1	2	4	-	F	replace 3 because it was used most recently (MRU).
5	1	1	2	4	-	H	page is already present in memory.
6	2	1	2	4	-	H	page is already present in memory.
7	5	1	5	4	-	F	replace 2 because it was used most recently (MRU).
8	1	1	5	4	-	H	page is already present in memory.
9	2	2	5	4	-	F	replace 1 because it was used most recently (MRU).
10	3	3	5	4	-	F	replace 2 because it was used most recently (MRU).
11	4	3	5	4	-	H	page is already present in memory.
12	5	3	5	4	-	H	page is already present in memory.

Final calculation: Page faults = 12 - 5 = 7. Hit ratio = 5/12 × 100 = 41.67%. Miss ratio = 7/12 × 100 = 58.33%.

Optimal

Optimal – 3 Frames: Step-by-Step Calculation

Total references = 12
Hits = 5
Page faults = 12 - 5 = 7
Hit ratio = 5/12 × 100 = 41.67%
Miss ratio = 7/12 × 100 = 58.33%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	1	2	4	-	F	replace 3 because its next use is farthest in the future (Optimal).
5	1	1	2	4	-	H	page is already present in memory.
6	2	1	2	4	-	H	page is already present in memory.
7	5	1	2	5	-	F	replace 4 because its next use is farthest in the future (Optimal).
8	1	1	2	5	-	H	page is already present in memory.
9	2	1	2	5	-	H	page is already present in memory.
10	3	3	2	5	-	F	replace 1 because it is not used again in the future (Optimal).
11	4	4	2	5	-	F	replace 3 because it is not used again in the future (Optimal).
12	5	4	2	5	-	H	page is already present in memory.

Final calculation: Page faults = 12 - 5 = 7. Hit ratio = 5/12 × 100 = 41.67%. Miss ratio = 7/12 × 100 = 58.33%.

QUESTION 2 – 4 FRAMES

FIFO

FIFO – 4 Frames: Step-by-Step Calculation

Total references = 12
Hits = 2
Page faults = 12 - 2 = 10
Hit ratio = 2/12 × 100 = 16.67%
Miss ratio = 10/12 × 100 = 83.33%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	1	2	3	4	F	an empty frame is available, so the page is loaded.
5	1	1	2	3	4	H	page is already present in memory.
6	2	1	2	3	4	H	page is already present in memory.
7	5	5	2	3	4	F	replace 1 because it entered memory first (FIFO).
8	1	5	1	3	4	F	replace 2 because it entered memory first (FIFO).
9	2	5	1	2	4	F	replace 3 because it entered memory first (FIFO).
10	3	5	1	2	3	F	replace 4 because it entered memory first (FIFO).
11	4	4	1	2	3	F	replace 5 because it entered memory first (FIFO).
12	5	4	5	2	3	F	replace 1 because it entered memory first (FIFO).

Final calculation: Page faults = 12 - 2 = 10. Hit ratio = 2/12 × 100 = 16.67%. Miss ratio = 10/12 × 100 = 83.33%.

LRU

LRU – 4 Frames: Step-by-Step Calculation

Total references = 12
Hits = 4
Page faults = 12 - 4 = 8
Hit ratio = 4/12 × 100 = 33.33%
Miss ratio = 8/12 × 100 = 66.67%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	1	2	3	4	F	an empty frame is available, so the page is loaded.
5	1	1	2	3	4	H	page is already present in memory.
6	2	1	2	3	4	H	page is already present in memory.
7	5	1	2	5	4	F	replace 3 because it was used least recently (LRU).
8	1	1	2	5	4	H	page is already present in memory.
9	2	1	2	5	4	H	page is already present in memory.
10	3	1	2	5	3	F	replace 4 because it was used least recently (LRU).
11	4	1	2	4	3	F	replace 5 because it was used least recently (LRU).
12	5	5	2	4	3	F	replace 1 because it was used least recently (LRU).

Final calculation: Page faults = 12 - 4 = 8. Hit ratio = 4/12 × 100 = 33.33%. Miss ratio = 8/12 × 100 = 66.67%.

MRU

MRU – 4 Frames: Step-by-Step Calculation

Total references = 12
Hits = 6
Page faults = 12 - 6 = 6
Hit ratio = 6/12 × 100 = 50.00%
Miss ratio = 6/12 × 100 = 50.00%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	1	2	3	4	F	an empty frame is available, so the page is loaded.
5	1	1	2	3	4	H	page is already present in memory.
6	2	1	2	3	4	H	page is already present in memory.
7	5	1	5	3	4	F	replace 2 because it was used most recently (MRU).
8	1	1	5	3	4	H	page is already present in memory.
9	2	2	5	3	4	F	replace 1 because it was used most recently (MRU).
10	3	2	5	3	4	H	page is already present in memory.
11	4	2	5	3	4	H	page is already present in memory.
12	5	2	5	3	4	H	page is already present in memory.

Final calculation: Page faults = 12 - 6 = 6. Hit ratio = 6/12 × 100 = 50.00%. Miss ratio = 6/12 × 100 = 50.00%.

Optimal

Optimal – 4 Frames: Step-by-Step Calculation

Total references = 12
Hits = 6
Page faults = 12 - 6 = 6
Hit ratio = 6/12 × 100 = 50.00%
Miss ratio = 6/12 × 100 = 50.00%

Step	Page	F1	F2	F3	F4	H/F	What happened?
1	1	1	-	-	-	F	an empty frame is available, so the page is loaded.
2	2	1	2	-	-	F	an empty frame is available, so the page is loaded.
3	3	1	2	3	-	F	an empty frame is available, so the page is loaded.
4	4	1	2	3	4	F	an empty frame is available, so the page is loaded.
5	1	1	2	3	4	H	page is already present in memory.
6	2	1	2	3	4	H	page is already present in memory.
7	5	1	2	3	5	F	replace 4 because its next use is farthest in the future (Optimal).
8	1	1	2	3	5	H	page is already present in memory.
9	2	1	2	3	5	H	page is already present in memory.
10	3	1	2	3	5	H	page is already present in memory.
11	4	4	2	3	5	F	replace 1 because it is not used again in the future (Optimal).
12	5	4	2	3	5	H	page is already present in memory.

Final calculation: Page faults = 12 - 6 = 6. Hit ratio = 6/12 × 100 = 50.00%. Miss ratio = 6/12 × 100 = 50.00%.

QUESTION 2 – BELADY'S ANOMALY

FIFO with 3 frames:
Hits = 3
Page faults = 12 - 3 = 9
Hit ratio = 3/12 × 100 = 25.00%
Miss ratio = 9/12 × 100 = 75.00%

FIFO with 4 frames:
Hits = 2
Page faults = 12 - 2 = 10
Hit ratio = 2/12 × 100 = 16.67%
Miss ratio = 10/12 × 100 = 83.33%

Checking Belady's anomaly:
Page faults with 3 frames = 9
Page faults with 4 frames = 10
Increase in page faults = 10 - 9 = 1

Since 10 is greater than 9, the number of page faults increased when the number of frames was increased from 3 to 4. Therefore, **Belady's anomaly exists in FIFO for this sequence.**

QUESTION 2 – COMPARISON

Algorithm	3 Frames Hits	3 Frames Faults	3 Frames Hit Ratio	4 Frames Hits	4 Frames Faults	4 Frames Hit Ratio
FIFO	3	12 - 3 = 9	3/12 × 100 = 25.00%	2	12 - 2 = 10	2/12 × 100 = 16.67%
LRU	2	12 - 2 = 10	2/12 × 100 = 16.67%	4	12 - 4 = 8	4/12 × 100 = 33.33%
MRU	5	12 - 5 = 7	5/12 × 100 = 41.67%	6	12 - 6 = 6	6/12 × 100 = 50.00%
Optimal	5	12 - 5 = 7	5/12 × 100 = 41.67%	6	12 - 6 = 6	6/12 × 100 = 50.00%

Final Answer

For this sequence, FIFO shows Belady's anomaly because its page faults increase from 9 to 10 when the number of frames increases from 3 to 4.

For the given sequence, Optimal gives the theoretical minimum number of page faults. Among the practical algorithms compared here, MRU gives the best result for this particular sequence.

Final Results

Question	Frames	Algorithm	Hits	Page Faults	Hit Ratio	Miss Ratio
Q1	3	FIFO	6	19 - 6 = 13	6/19 × 100 = 31.58%	13/19 × 100 = 68.42%
Q1	3	LRU	5	19 - 5 = 14	5/19 × 100 = 26.32%	14/19 × 100 = 73.68%
Q1	3	MRU	7	19 - 7 = 12	7/19 × 100 = 36.84%	12/19 × 100 = 63.16%

Q1	3	Optimal	9	19 - 9 = 10	9/19 × 100 = 47.37%	10/19 × 100 = 52.63%
Q2	3	FIFO	3	12 - 3 = 9	3/12 × 100 = 25.00%	9/12 × 100 = 75.00%
Q2	3	LRU	2	12 - 2 = 10	2/12 × 100 = 16.67%	10/12 × 100 = 83.33%
Q2	3	MRU	5	12 - 5 = 7	5/12 × 100 = 41.67%	7/12 × 100 = 58.33%
Q2	3	Optimal	5	12 - 5 = 7	5/12 × 100 = 41.67%	7/12 × 100 = 58.33%
Q2	4	FIFO	2	12 - 2 = 10	2/12 × 100 = 16.67%	10/12 × 100 = 83.33%
Q2	4	LRU	4	12 - 4 = 8	4/12 × 100 = 33.33%	8/12 × 100 = 66.67%
Q2	4	MRU	6	12 - 6 = 6	6/12 × 100 = 50.00%	6/12 × 100 = 50.00%
Q2	4	Optimal	6	12 - 6 = 6	6/12 × 100 = 50.00%	6/12 × 100 = 50.00%